INCH-POUND

MIL-P-70533 (AR)
AMENDMENT 5
09 April 1993
SUPERSEDING
AMENDMENT 4
09 November 1992

MILITARY SPECIFICATION

PROJECTILE, 105MM, HE, RA, XM913/XM927
ROCKET PROPELLANT FOR

This amendment forms a part of Military Specification MIL-P-70533(AR), dated 29 October 1990, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Change "PROJECTILE" to "CARTRIDGE".
Change "XM913" to "M913" (and wherever it appears).

2.1.1 Add to SPECIFICATIONS, MILITARY the following:

"MIL-B-117 - Bags, Sleeves and Tubing"

Delete "MIL-C-70532 - Cartridge....Packing".

PAGE 2

2.1.1 Add to STANDARDS, MILITARY the following:

"MIL-STD-129 - Marking for Shipment and Storage"

PAGE 3

Inhibitor peel strength: Delete first sentence and replace with new requirement:

"The average peel strength of the inhibitor/propellant bond at 70 ± 5 F shall be 20 pounds per linear inch (pli) minimum. No individual sample tested shall exhibit a peel strength less than 14 pli during acceptance testing."

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* Table II, Delete in its entirety and replace with new Table II:

"Table II. Mechanical Properties of Propellant

Temp. (±5F)	Characteristic	Avg. of 5 readings (minimum)	Single Sample readin (minimum)
-50	Ultimate Strain	16.5%	14.5%
"	Modulus of Elasticity	11,000 PSI	10,000 PSI
"	Ultimate Stress	455 PSI	440 PSI
72	Ultimate Strain	21 %	17 %
"	Modulus of Elasticity	2,000 PSI	1,800 PSI
"	Ultimate Stress	180 PSI	170 PSI
"	Strain at Max. Stress	20 %	16 %
145	Ultimate Strain	14.4%	13 %
"	Modulus of Elasticity	1,600 PSI	1,450 PSI
"	Ultimate Stress	139 PSI	135 PSI

3.9.1 Delete in its entirety.

3.10 Add: "(First Article only)" after title.

PAGE 8

Propellant after mixing and curing. Change the conformance criteria for heat of reaction from "5 each" to "2 each".

Complete Rocket Grain. Delete "4.5.5/4.4.2.4" in 2 places and substitute "4.5.6/4.4.2.3".

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4.4.2.3 Change the inspection method reference of special 1 from "4.5.5." to "4.5.6". Change the conformance criteria of major 106 from "3.12" to "Level III".

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Add the following:

"Major				
109	Weight of grain (3 segments)	100%	3.2	Scale
Minor				
202	Inhibitor flash (excessive)	Level V	3.7.3.1	Visual"

PAGE 14

4.5.1 Strand burning characteristics of propellant. Change the last sentence of the paragraph to read as follows:

"Three strands will be tested at each of the temperature, pressure conditions in Table I of paragraph 3.6, and the average of the three values shall be used for compliance with the requirements."

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4.5.2 Mechanical properties of propellant. Delete the last sentence and replace with the following new sentence:

"If either an individual sample or five-sample average fails to meet the requirement, the batch of propellant the samples represent shall be rejected."

4.5.2 Mechanical properties of propellant. Add the following new subparagraph:

"4.5.2.1 Retest. If the specimens taken from the bulk sample fail to meet the requirement, the contractor may retest for all mechanical properties using randomly selected grains, or segments, as applicable, representative of the failed bulk sample batch. The specimen shall be the largest attainable to yield a tensile coupon which conforms to the requirements of 4.5.2, with the exception of the tab end length."

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4.5.4 Delete in its entirety and substitute the following new paragraph:

"4.5.4 *Inhibitor peel strength*. Each batch of inhibitor will be tested with each batch of propellant that it is utilized with. A representative sample of each batch of inhibitor material or part thereof shall be bonded to a representative sample from the propellant batch to which it is bonded in a finished grain. This sample shall be cured in the same curing chamber under the same conditions for the same length of time and be tested in accordance with section 4.7.5.4 of CPIA Pub. 21. Two test specimens shall be tested for each batch of propellant. If any specimen fails to meet the specified requirements the individual propellant/inhibitor combination material batch represented shall be rejected."

4.5.5 Tool control dimensions. Delete in its entirety.

PAGE 16

4.5.7 Change "XM229 series propelling charge" to "PXR 200 propelling charge" in the first sentence of the second paragraph.

PAGE 17

5.1 Add the following after last sentence: "Appropriate cushioning and partitions shall be provided to protect the items from physical damage. Sufficient fillers must be used to insure a tight pack. In addition, adequate protection to damage from moisture must be provided. (Enclosing a number of items in a sealed barrier bag, meeting the requirements of Type I or II, Class E, Spec MIL-B-117 would satisfy this requirement.)"

5.2 Add the following: "5.2 Marking. The items, packaged as in 5.1 shall be marked in accordance with MIL-STD-129 and Code of Federal Regulations (CFR), Title 49, Parts 100-199."

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